

SUIII! A Ball Focus and Identity Memory for Robust Soccer Player Tracking

Shih-Wen Liu
National Cheng Kung University
Tainan, Taiwan
ne6134079@gs.ncku.edu.tw

Wei-Ta Chu
National Cheng Kung University
Tainan, Taiwan
wtchu@gs.ncku.edu.tw

Introduction



Goal: Improving ID assignment in multi-player soccer videos under occlusion and crowding match.

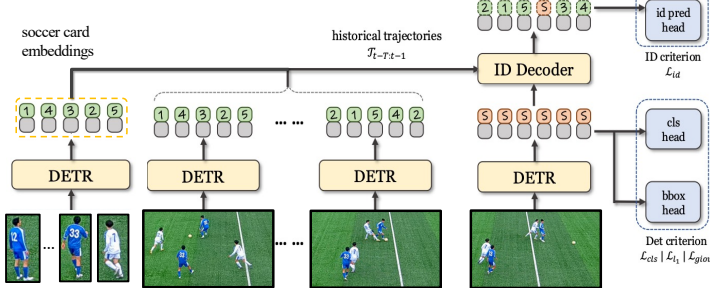
Method: Our framework leverage group motion dynamics (“ball focus”) and long-term appearance memory (“soccer card”) to stabilize tracking.

1. SUIII! Framework

We proposed 2 main concepts to track soccer players:

Ball Focus: a Particle Swarm Optimization (PSO)-inspired token that estimate the group’s motion center, which drives attention to the critical point for more informed tracking outcomes.

Soccer Card Memory: ID prototypes that retain each player’s clearest appearance embedding that tackles occlusions.



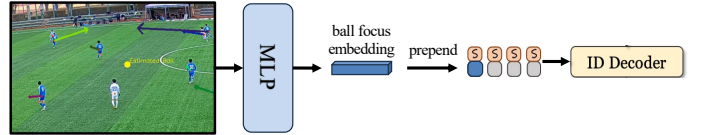
2. Ball Focus

Critical Point Estimation:

We first compute each player’s instantaneous motion using positions at frames $Frame_{t-60}$, $Frame_{t-30}$, and $Frame_{t-1}$. From these we derive velocity v_i and acceleration a_i for player i . By projecting the line together, we obtain the critical point c_i at $Frame_{t-1}$.

Ball Focus Token:

The 2D point c_i together with each player’s v_i and a_i are concatenated and passed through a lightweight MLP to produce a D -dimensional feature vector. This ball focus token encodes both the group’s common goal and individual motion cues, and is prepended to the ID decoder’s input sequence.



Motion Bias:

Ball Focus Token conditions the the detection head’s query features, induces a motion bias that steers bounding-box refinement toward the group’s common goal.

3. Soccer Card Memory

For each track ID, we maintain a prototype embedding representing the clearest appearance observed so far. At each frame, the current ID embedding is scored by a learnable-weighted fusion of detection confidence, box size ratio, cosine-similarity, speed, and acceleration:

$$s_i(t) = \alpha \cdot [c_i, r_i, \sigma_i, \|v_i\|, \|a_i\|]^T$$

If the new clarity score exceeds the stored score, we update the memory bank with new embedding.

Experiments

1. Comparison with SOTAs

Dataset:

SoccerTrack from ACM MMSPORTS 2025 with 10 soccer matches, each video contains roughly 6000 frames. 7 for Training, 1 for Validation and 2 for Testing.

Metrics:

- HOTA: Tracking and Detection by computing the geometric mean of DetA and AssA (balanced).
- DetA: Pure Detection.
- AssA: Pure Tracking.
- MOTA: Tracking and Detection (more on detection).
- IDF1: focusing purely on ID correctness.

Model	HOTA	DetA	AssA	MOTA	IDF1
SoccerTrack	29.9	50.2	18.0	59.3	32.0
MOTIP (base model)	28.5	57.5	14.3	65.9	27.7
+ Motion bias					
+ Ball Focus token	30.0	57.4	15.7	66.6	29.8
+ Soccer Card	29.5	57.5	15.3	66.1	28.9
Ours	30.1	57.4	15.8	66.6	30.0

Table 1. Comparison of Tracking Methods on Soccer Dataset

2. Visualization

Top figure: Speed vectors for each player with the estimated critical point highlighted (yellow).

Bottom figure: MOT results illustrating stable ID tracking using Ball Focus and Soccer Card Memory.

